

NLP Applications-Assignment1

Part B - Literature Survey

Topic: Knowledge Graph Construction

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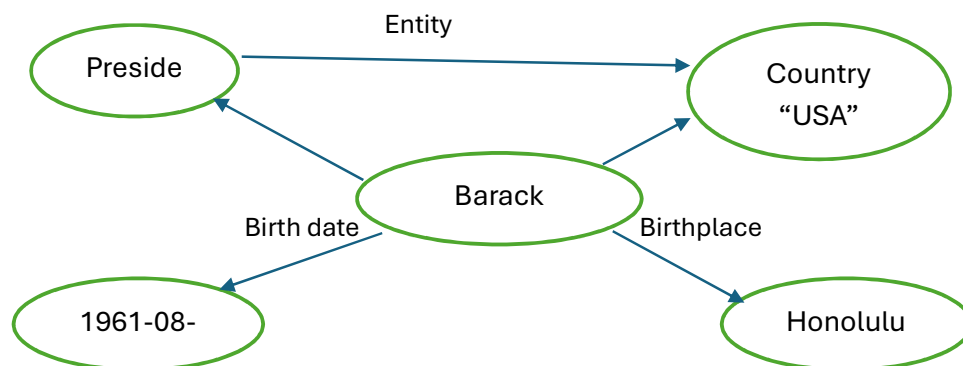
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1. Introduction

A Knowledge Graph (KG) in Natural Language Processing (NLP) is a structured representation of information that captures entities and their relationships in a graph format. It enhances understanding, reasoning, and retrieval in NLP tasks.

A Graph based data structure where:

- Nodes represent entities (e.g., 'Barack Obama', 'USA', 'President').
- Edges represent relationships (e.g., Barack Obama → President of → USA).
- Knowledge graphs store semantic relationships and enable reasoning beyond keywords.



Knowledge graphs help NLP systems:

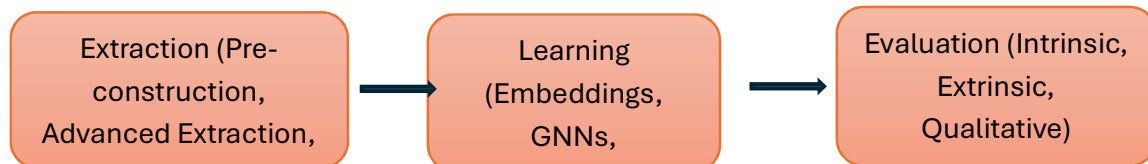
- Disambiguate meaning (e.g., 'Apple' as a company vs. fruit)
- Improve search and question answering by linking concepts

- Enable reasoning for tasks like recommendation systems or dialogue systems

2. Literature Review

Recent literature (2022-2024) highlights phases: Extraction, Learning, and Evaluation. Extraction involves entity and relation identification from text, images, and multi modal data. Learning focuses on embeddings, GNNs, and Transformer-based models. Evaluation includes intrinsic, extrinsic, and qualitative metrics.

Taxonomy of Knowledge Graph Construction



2.1. Extraction and Integration Approaches

Researchers have developed a variety of methods to extract entities, relations, and structure from different data types and integrate them into Knowledge Graph.

2.1.1. Principle of Extraction Approaches

a) KG Construction

- **Entity & Relation Extraction:** Using NER and dependency parsing.
- **Integration with LLMs:** Recent work explores combining KGs with large language models to reduce hallucinations and improve factuality.
- **End-to-End Models:** Models like REBEL perform joint entity and relation extraction for KG building from text.

b) KG Representation & Learning

- **Embedding Techniques:** Linear/bilinear models, factorization, and neural networks for KG representation.
- **Graph Neural Networks (GNNs)** and **Transformers** are increasingly used for KG learning and integration with NLP tasks.

c) Integration with NLP

- **Knowledge-Grounded Dialogue & QA:** Injecting structured knowledge into LLMs for better reasoning.
- **Retrieval-Augmented Generation (RAG):** Popular for grounding LLM outputs in factual data

2.2 Extraction Approaches along with Strengths & Limitations

2.2.1 Rule-based & Statistical NLP Pipelines

What they do: Use classical NLP techniques rule-based extraction, pattern matching, statistical methods often relying on structured or semi-structured data (e.g. relational databases, XML/JSON, semi-structured text).

Early/enterprise-oriented KGC used structured sources (databases, spreadsheets) or semi-structured data (tables, XML/HTML) and applied parsing + mapping + schema alignment to integrate data into KG.

(<https://john.mccr.ae/papers/masoud2021automatic.pdf>)

For text-based extraction: Named Entity Recognition (NER), basic Relation Extraction (RE), and linking to ontology or canonical entities. In simpler domains or small-scale corpora, even manual/entity-centric extraction has been used.

Strengths:

- Transparent and interpretable.
- When data is well-structured or semi-structured, mapping to KG is straightforward and reliable.

Limitations:

- Fragile with noisy/unstructured text.
- Hard to scale manual rules don't generalize well across domains.
- Difficult to handle ambiguous language, synonyms, varying contexts, entity disambiguation, coreference etc.

Because of these limitations and the explosion of unstructured data, newer methods have emerged.

2.2.2. Machine-Learning & Deep Learning based Extraction

As NLP advanced, KGC began to adopt ML/DL techniques to automatically extract entities/relations from unstructured text, and integrate them more robustly. Key components:

NER + Relation Extraction (RE) + Entity Linking / Disambiguation / Normalization: identifying entity mentions, their types; extracting relations between them; linking mentions to canonical entities (or external knowledge bases). (<https://link.springer.com/article/10.1186/s40537-023-00774-9>)

Use of modern NLP models (e.g. transformers, BERT variants) combined with sequence modeling (CRF, BiLSTM-CRF) or CNN/IDCNN layers to improve recognition on long, domain-specific, or noisy text.

For certain domains (e.g. geoscience, biomedical), domain-adapted models have been built to capture domain-specific terminology, relationships, and semantics.

Strengths:

- Much better handling of unstructured natural language.
- Can generalize across varying text styles, domains (with fine-tuning/transfer learning).
- More scalable than manual/rule-based approaches.

Limitations:

- Requires annotated data or knowledge bases for training (data intensive).
- Performance may drop in low-data or highly specialized domains.
- Entity disambiguation/entity linking remains challenging, especially when context is ambiguous or domain specific.
- Integration/fusion across data from multiple sources remains non-trivial (heterogeneous formats, inconsistencies, noise).

An example: in a recent geological-domain KG construction task, researchers compared different models including BERT-based BiLSTM-IDCNN-CRF, showing improved entity & relation extraction from long geological texts.

2.3.3 Hybrid / Machine-Learning-Guided Knowledge Graph Construction

A growing trend is **hybrid** pipelines: combine structured data, unstructured textual data, and sometimes domain ontologies using ML + rules + human-in-the-loop to maximize coverage and accuracy.

A recent survey on ML-guided KG construction organizes methods along **entity learning**, **ontology/schema learning**, and **knowledge reasoning/learning**. (<https://www.techscience.com/CMES/v139n1/55119>)

For example: after extracting entity and relation candidates (via NLP/ML), systems perform **entity linking/alignment**, **schema (ontology) alignment or enrichment**, integrate data from different sources, and perform **consistency checks** / **refinement** / **fusion**. (<https://www.techscience.com/CMES/v139n1/55119>)

For structured + unstructured data integration: some enterprise-level pipelines take data from databases, textual documents, and other sources map them to a unified schema, then perform merging/alignment to build a consolidated KG.

Strengths:

- Broader coverage can tap into multiple data modalities.
- Leverages strengths of both rule-based/structured methodologies and ML-based flexibility.

- More robust integration and better overall KG quality if designed well.

Limitations & Challenges:

- Complexity: building such pipelines is heavy engineering work.
- Data heterogeneity: different sources have different schema/semantics merging and aligning them is hard.
- Inconsistent or conflicting information may require human or high-level reasoning to resolve.
- Need for ontology/schema design which can become a bottleneck, especially for domain-specific or evolving domains.

2.3.4. Emerging: LLM-Empowered & Generative Approaches

With the recent advances in large language models (LLMs) and generative models, a new paradigm for KG construction is emerging. Instead of (or in addition to) rule-based or ML-based extraction pipelines, generative or LLM-based systems are used to produce structured knowledge.

A recent survey titled LLM-empowered knowledge graph construction: A survey (2025) reviews how LLMs reshape classical KG pipelines (ontology engineering, extraction, fusion). (<https://arxiv.org/abs/2510.20345>)

Another work, Generative Knowledge Graph Construction: A Review, explores methods where the task of KG construction is cast as a “structure prediction” problem: given raw input (such as sentences), a model predicts structured output (entities, relations, events) directly.

These methods often use pre-trained LLMs fine-tuned or prompted to extract triples (entity, relation, entity), link entities, and even reason about relations implicitly.

(<https://aclanthology.org/2022.emnlp-main.1.pdf>)

Strengths:

- Very flexible: can adapt to new domains with minimal engineering (especially schema-free methods).
- Good for domains with limited annotated data (few-shot / zero-shot capabilities).
- Potentially can capture more abstract, conceptual knowledge beyond literal text mentions.
- Reduces need for heavy manual pipeline engineering.

Limitations / Risks:

- Accuracy and consistency can be problematic generative models may hallucinate facts, produce inconsistent triples.

- Hard to guarantee ontology/schema compliance, canonicalization, entity disambiguation.
- Quality control / validation become critical and nontrivial.
- For use in sensitive domains (medicine, law, etc.), verifying correctness is difficult.

2.3.5. Domain-Specific & Multimodal Graphs Handling Heterogeneous Sources

Many real-world applications need KGC from not just plain text, but **multimodal** data (text, images, structured databases, domain-specific formats) and **domain-specific data** (e.g. biomedical, geoscience, enterprise records, EHR).

A recent survey on healthcare KG construction highlights that typical pipelines use NER, Named Entity Disambiguation (NED) / Entity Linking (NEL) to extract entities from heterogeneous sources (clinical notes, EHRs, structured records) and then integrate them appropriately. (<https://link.springer.com/article/10.1186/s40537-023-00774-9>)

In geoscience / geological domain: researchers have used advanced NLP models to extract domain-specific entities/relations from geological reports dealing with many domain-specific terminologies, context dependencies, and long-form text. (<https://www.sciencedirect.com/science/article/pii/S0169136825000745>)

There is growing interest in **multimodal knowledge graphs** (e.g. integrating images + text/unstructured data) though research is relatively limited. A survey on entity alignment notes that multimodal KGs (text + images) enable richer representation and better cross-modal alignment. (<https://link.springer.com/article/10.1007/s10462-024-10866-4>)

Implications:

For domain-specific KG construction, it's often necessary to customize extraction pipelines (domain-adapted NER/RE models), ontology/schema, and data integration strategies.

Multimodal or cross-source integration remains challenging normalization, canonicalization, resolving duplicates or contradictory information, linking across modalities.

Hybrid or LLM-based approaches are becoming attractive for such complex, heterogeneous data environments.

Comparative Table of Extraction Approaches

Extraction Approach	Accuracy	Scalability	Domain Adaptability	Automation
Rule-Based	High	Moderate	Moderate	Low
ML-Based	Moderate	High	Moderate	Moderate
LLM-Assisted	High	High	High	High
Graph-Based	High	Moderate	Moderate	Moderate
Multimodal	High	Moderate	High	Moderate

3. Integration of Information from Multiple Sources

Integration strategies include ontology alignment, schema mapping, and entity resolution across heterogeneous datasets. Advanced methods leverage self-supervised learning and contrastive techniques for robust alignment.

3.1. Entity Extraction (NER – Named Entity Recognition)

Goal: Identify entities (people, organizations, locations, dates, products, etc.) from unstructured text.

Working Method:

- **Input:** Raw text (e.g., “Microsoft acquired LinkedIn in 2016 for \$26 billion.”)
- **Output:** Tagged entities:
 - Microsoft → Organization
 - LinkedIn → Organization
 - 2016 → Date
 - \$26 billion → Monetary Value

Techniques:

- **Rule-based:** Regex patterns for dates, currency, etc.
- **Statistical / ML models:** CRF, HMM
- **Deep Learning:** BiLSTM-CRF, Transformer-based models (BERT, spaCy, HuggingFace)

Tools: spaCy, NLTK, Stanford NER, HuggingFace Transformers.

3.2. Relation Extraction

Goal: Detect semantic relationships between entities (e.g., “Microsoft **acquired** LinkedIn”).

Types of relations:

- **Binary relations:** Company-Acquired-Company
- **N-ary relations:** Person-Founded-Company-In-Year

- **Hierarchical relations:** is used in ontologies and knowledge graphs to represent **class-subclass** or **whole-part** structures.

Approaches:

- **Pattern-based:** Dependency parsing + linguistic rules.
- **Supervised ML:** Train on labelled relation datasets.
- **Deep Learning:** Use BERT for relation classification.

3.3 Ontology Design

Goal: Define a schema for entities and relationships (the backbone of a Knowledge Graph).

Components:

- **Classes (Entity Types):** Person, Organization, Location, Event
- **Properties:** Name, Date, Revenue
- **Relations:** worksFor, locatedIn, acquired

Standards: RDF, OWL, SKOS **Tools:** Protégé (for ontology modelling), Neo4j (graph DB), RDF triple stores.

3.4 Linking & Integration

Goal: Connect extracted entities to existing knowledge bases (e.g., Wikidata, DBpedia).

Steps:

- **Entity Linking:** Match “Microsoft” → Q2283 in Wikidata.
- **Disambiguation:** Handle cases like “Apple” (fruit vs company).
- **Integration:** Merge into Knowledge Graph with unique IDs.

Techniques:

- String similarity + context-based disambiguation.
- Use APIs like Wiki data SPARQL, Bedia Lookup.

Popular Knowledge Graphs:

- Google Knowledge Graph
- Wikidata
- DBpedia
- ConceptNet

3.5 Advanced Integration Approaches

a) Self-Supervised Learning

- **Concept:** Generate pseudo-labels or contrastive pairs without manual annotation.
- **Example:** Contrastive learning for entity alignment maximizes similarity for positive pairs (same entity) and minimize for negatives.

b) Contrastive Techniques

- **How It Works:**
 - Represent entities as embeddings.
 - Apply **contrastive loss** to align similar entities across graphs.
- **Benefits:**
 - Handles **data sparsity**.
 - Improves robustness in multilingual or cross-domain settings.

4. Applications

- **Question Answering (QA):** KG-based QA systems outperform text-only models in factual accuracy
- **Semantic Search:** Improves relevance by leveraging entity relationships
- **Healthcare & Biomedical NLP:** Ontology-grounded QA for rare diseases
- **Enterprise Knowledge Management:** Enhances decision support and recommendation systems
- **Chatbots:** Context-aware responses using KG
- **Engines Recommendation:** Suggest items based on graph connections

5. Challenges

Despite progress, many challenges remain especially when dealing with **heterogeneous or noisy sources**:

- **Scalability:** Handling large, heterogeneous data sources
- **Entity Disambiguation:** Resolving ambiguities in names and references
- **Dynamic Updates:** Keeping KGs current with real-world changes
- **Evaluation:** Lack of standardized benchmarks for KG quality and integration performance
- **Fusion with LLMs:** Addressing hallucinations and ensuring interpretability in KG-augmented models

6. Overview

Knowledge Graphs represent entities and their relationships in a graph structure, enabling semantic understanding and reasoning. Since Google introduced its KG in 2012, research has surged, especially in NLP tasks like question answering, semantic search, and dialogue systems. A systematic survey rapid growth post-2017 and diversification across domains such as healthcare, social media, and scholarly research.

7. Emerging Trends

- **KG + LLM Synergy:** Mitigating hallucinations and improving reasoning
- **Multimodal KG Construction:** Combining text, image, and audio for richer graphs
- **Domain-Specific KGs:** Growth in specialized areas like medicine and finance
- **Explainability & Ethics:** Ensuring fairness and transparency in KG-driven NLP systems

Conclusion

The field of Knowledge Graph Construction is **rapidly evolving**: from early rule- or pattern-based pipelines anchored in structured data, to modern ML/DL-based extraction from unstructured texts, and now to emerging **LLM-based, generative, hybrid pipelines** capable of integrating heterogeneous, multimodal sources.

While current techniques have made KG construction more scalable and flexible, significant challenges remain especially around **integration, canonicalization, consistency, evaluation, and evolution of knowledge**. Research is increasingly focusing on hybrid and dynamic KG pipelines, domain-specific adaptation, and leveraging large pre-trained language models for more flexible knowledge discovery.

For a researcher or practitioner: a **hybrid approach** (mixing extraction, schema/ontology design, ML/NLP, and human-in-the-loop validation) remains the most practical and reliable path for now especially when building KGs from diverse, noisy, or domain-specific data sources.